

Lineage-Primary Preservation: A Model for Digital Longevity

Author: Matthew John Keen

The Wedding Ring Analogy (Personal)

I bought two identical wedding rings.

One was intended to be worn every day by my wife. The other was placed carefully in storage, kept pristine as a backup.

At the time, the reasoning was practical. If the worn ring were ever lost or damaged beyond repair, the untouched ring could replace it exactly. Both were materially identical; only their histories would differ.

Over time, that difference became decisive.

The Worn Ring (The Lineage-Primary Object)

The ring my wife wears every day is the authentic wedding ring. It has accumulated scratches and marks through ordinary life. It has aged alongside the marriage itself.

Those changes do not diminish its meaning. On the contrary, they are the evidence of continuity. The ring's authenticity does not come from its original condition, but from its uninterrupted role as the signal of a specific event, carried forward through time.

In lineage-primary terms, this ring is the primary carrier of identity. The change it undergoes is not corruption, but acceptable drift within recognisable bounds.

The Stored Ring (The Bit-Checked Reference)

The second ring remains unused, unchanged, and materially perfect. It functions as a reference copy—a stable point against which the wear of the primary ring can be understood.

Yet it is not authoritative. If the stored ring were lost, the marriage would not lose its continuity. If the worn ring were replaced entirely by the pristine one, something essential would be broken: the lived lineage would be severed.

The preserved object would remain intact, but the history would not.

Rethinking Digital Preservation

The digital preservation community has long operated under an assumption inherited from early computing: that success is defined by exact, bit-level fidelity. Any deviation from the original file state is treated as failure.

While effective over short timescales, this fixity-first paradigm risks cliff-edge failure over centuries. Files may remain technically intact yet become uninterpretable as technical, institutional, or cultural contexts disappear.

A lineage-primary model reorders this relationship between data and history. Rather than attempting to freeze digital objects in place, it prioritises the sustained legibility of identity over time.

Core Properties of the Lineage-Primary Model

Identity Through Transformation

Identity is carried forward through a provenance graph that records every migration, repair, or reconstruction as a lineage event.

Graceful Degradation

Partial loss or institutional disruption does not end historical continuity. The loss of a single instance does not invalidate identity.

Species Descriptors

Semantic intent and permissible variation are bounded to prevent uncontrolled drift.

Defense Against Synthetic Media

Authenticity is tied to documented ancestry rather than appearance. Synthetic replicas cannot claim false historical origin without lineage.

Conclusion

By repositioning fixity as reference rather than authority, lineage-primary preservation offers a more resilient approach to long-term survival. It accepts that artifacts—digital or physical—must be allowed to age to endure.

Preservation becomes the maintenance of historical truth, not the denial of change.